COMPARATIVE ANALYSIS OF DEEP LEARNING MODELS FOR SKIN CANCER CLASSIFICATION

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PROBLEM STATEMENT

- The project aims to implement machine learning algorithms to classify dermoscopic images from the HAM10000 dataset into seven distinct skin lesion categories.

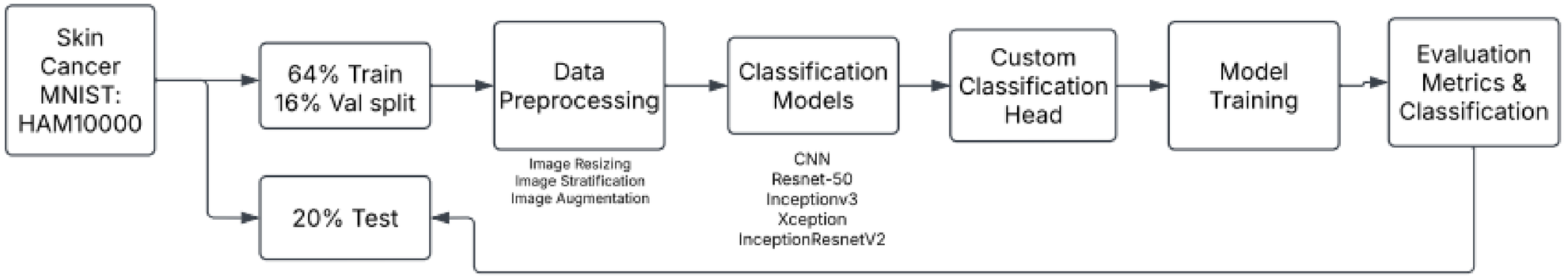
- Additionally, it involves a comparative analysis of multiple deep learning models to identify the one that delivers the most accurate, reliable, and generalizable performance under constrained, real-world conditions.

MOTIVATION

- Skin cancer is one of the most common and potentially life-threatening cancers worldwide.
- Early and accurate diagnosis plays a critical role in improving patient outcomes. However, traditional methods are time-consuming, expertise-dependent, and prone to human error.
- With the growing success of deep learning in medical imaging, there is a strong need to explore automated classification techniques that can assist dermatologists in making faster and more reliable diagnoses.
- This project is motivated by the potential of Convolutional Neural Networks (CNNs) to accurately classify complex dermoscopic images, and by the opportunity to evaluate how different architectures perform on imbalanced, real-world medical data.

METHODS

Project Architecture



METHODS

Data Preprocessing

Deduplication
- Removed duplicate images to avoid data leakage.
- Dataset reduced from 10,015 to 7,470 unique images.

Train-Validation-Test Split
- Stratified split to maintain class balance:
- 64% training, 16% validation, 20% testing.

Label Encoding
- Converted class names (e.g., mel, nv) into numerical labels using LabelEncoder.

Image Resizing
- Images resized based on model requirements:
- 128×128 for CNN
- 224×224 for ResNet-50
- 299×299 for InceptionV3, Xception, and InceptionResNetV2

Normalization
- CNN: normalized to [1, 1]
- Other models: normalized to [0, 1] to match pretrained ImageNet expectations

Data Augmentation (Training Set Only)
- Applied random rotations, flips, shifts, and zooms

METHODS

Models Used

- Custom CNN
  - A simple 4-layer convolutional neural network built from scratch.
  - Each Conv2D block is followed by Batch Normalization, ReLU activation, and MaxPooling.
  - A Global Average Pooling layer and a Dense classifier with Dropout (0.5) prevent overfitting.
  - Trained using Adam optimizer and categorical cross-entropy loss.

- ResNet-50 (PyTorch)
  - Pretrained ImageNet Model.
  - All layers except the final classification head, which was replaced with a custom fully connected layer to output predictions for 7 classes.
  - Trained using CrossEntropyLoss and Adam optimizer with a learning rate of 0.001.

METHODS

Models Used

- InceptionV3
  - A deep convolutional network from Google that uses Inception modules to capture multi-scale features efficiently.
  - Pretrained ImageNet Model with a Custom head: GlobalAvgPooling → BatchNorm → Dropout → Dense(512) → Dropout → Dense(7)
  - Optimizer: Adam (1e-4), Loss: Sparse Categorical Crossentropy
  - Early stopping based on validation accuracy

- Xception
  - An advanced CNN architecture that replaces Inception modules with depthwise separable convolutions for improved speed and performance.
  - Pretrained on ImageNet with same custom classification head and parameters as InceptionV3

- InceptionResNetV2
  - Combines the power of Inception blocks with residual connections to enable deeper, stable networks.
  - Pretrained on ImageNet with same custom classification head and parameters as InceptionV3

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Evaluation Metrics

- Accuracy reflects the overall correctness of predictions by measuring the proportion of correctly classified images out of the total.
- Precision focuses on the number of positive predictions that were actually positive minimizing false alarms.
- Recall, on the other hand, is the ratio of the correctly predicted positive instances to the total positive instances.
- F1 score is the harmonic mean between both precision and recall, offering a single metric that is especially useful when dealing with imbalanced classes.

EXPERIMENTAL RESULTS AND ANALYSIS

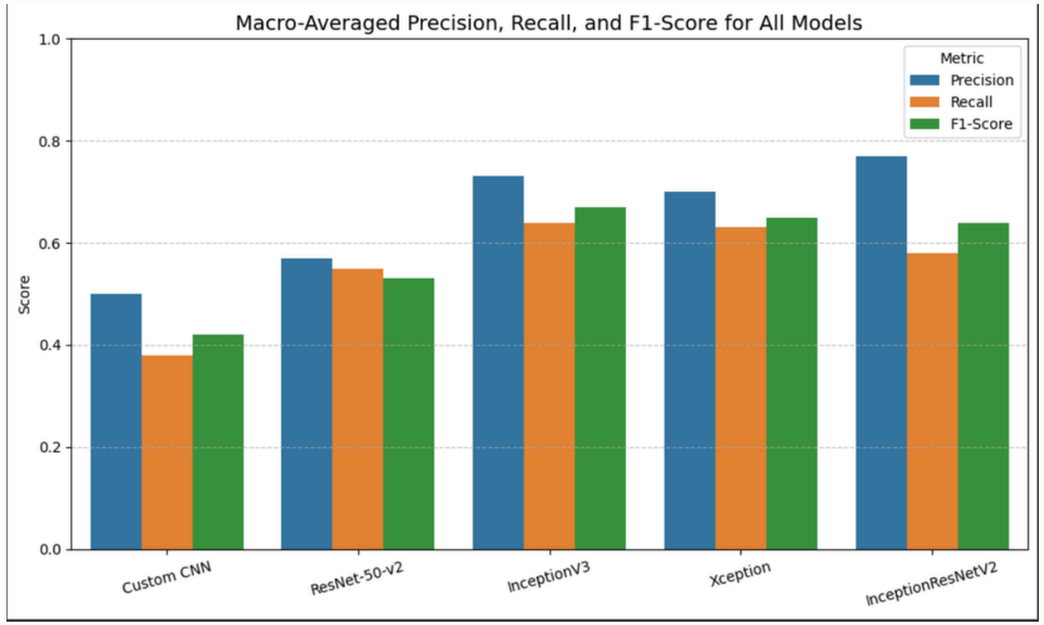
- Xception achieved the highest accuracy (85.63%), making it the best at overall predictions.
- InceptionV3 scored the highest macro Fl-score (0.67), showing stronger class-wise balance, especially helpful for rare lesion types.
- InceptionResNetV2 maintained high performance in both metrics, reflecting a good trade-off between depth and precision.
- ResNet-50, had a lower macro Fl (0.53), indicating challenges in identifying minority classes despite decent accuracy (75%).
- Custom CNN underperformed with both metrics (74% accuracy, 0.42 macro Fl), struggling particularly with underrepresented and visually ambiguous lesions.

Model	Accuracy	Macro F1-Score
Custom CNN	74.00	0.42
Resnet-50-v2	75.00	0.53
Inception-v3	84.00	0.67
Xception	85.63	0.65
InceptionResnetv2	83.62	0.64

EXPERIMENTAL RESULTS AND ANALYSIS

Macro Averaged Precision, Recall and F1 Scores

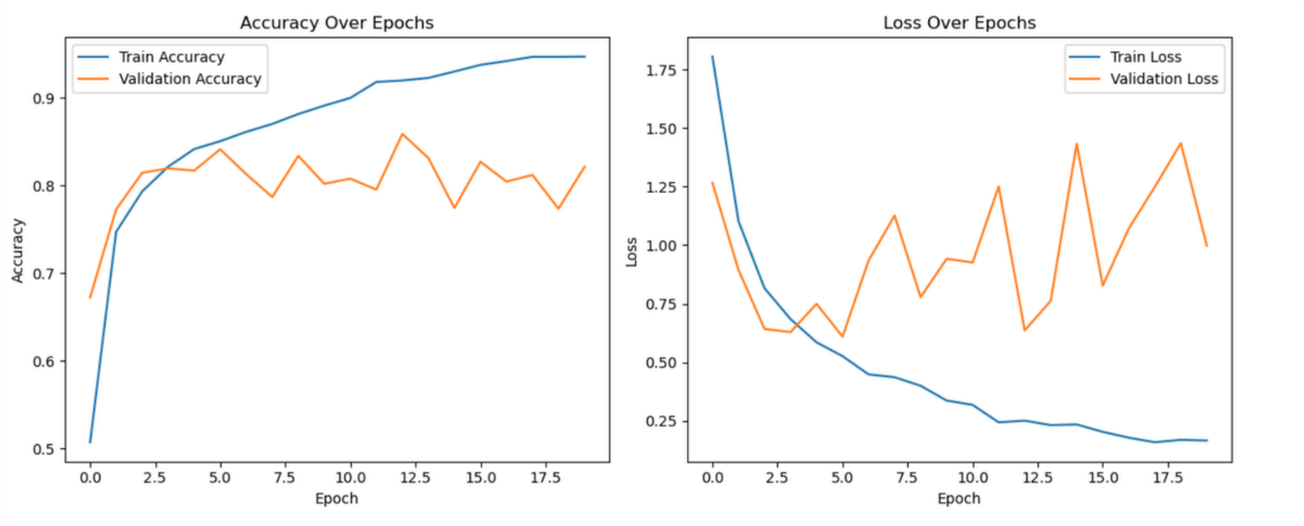
- InceptionV3: Caught the most actual cancer cases (highest recall) indicating it was most effective at correctly identifying true positive skin cancer cases
- InceptionResNetV2: Has the highest precision and made the fewest false cancer predictions— useful if you want to avoid wrongly telling patients they might have cancer.
- Xception: Balanced both precision and recall, it caught most cancers classes and didn't make too many wrong calls, making it good for general hospital use.
- ResNet-50-v2: Made more incorrect cancer predictions compared to the top models but better than Custom CNN. It's still not ideal.
- Custom CNN: Has low Precision, recall and F1 scores. Missed a lot of actual cancers (low recall) and is unsafe for skin cancer detection, might let harmful lesions go unnoticed.



EXPERIMENTAL RESULTS AND ANALYSIS

Best model: Xception Training and Validation Curves

- Training accuracy steadily increased to ~94%, while validation accuracy remained stable around 80–84%.
- Training loss consistently dropped, but validation loss showed fluctuations suggesting some overfitting risk.
- Despite this, Xception maintained strong validation performance, confirming its robustness and generalization ability.
- The best Xception model (based on highest validation accuracy) was automatically saved using model checkpointing,



CONCLUSION AND FUTURE WORK

- Among the five models evaluated, transfer learning architectures like Xception and InceptionV3 delivered the most reliable performance, especially on this imbalanced skin cancer dataset.
- Simpler models like ResNet50 and Custom CNN struggled with underrepresented classes making them less suitable for high-stakes diagnostic use.
- Overall, deep CNN-based transfer learning models show significant promise in classifying dermatoscopic images, showing promise in real-world settings.
- Improved Minority Class Detection: Future efforts should focus on enhancing sensitivity for underrepresented but clinically critical classes

THANK YOU